



Sentimental-Behavioral Debt Recovery

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MONITORED ACCOUNTS

30,000

Credit pool \$5.02B

HIGH-RISK ACCOUNTS

4,345

T4 + T5 · \$0.44B exposure

PORTFOLIO DISTRESS

0.51

Elevated

BEHAVIOURAL ANOMALIES

1,500

5.0% of portfolio · BiLSTM P95

Overview

Portfolio

Customer Insight

Audit & Risk

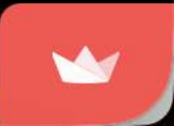
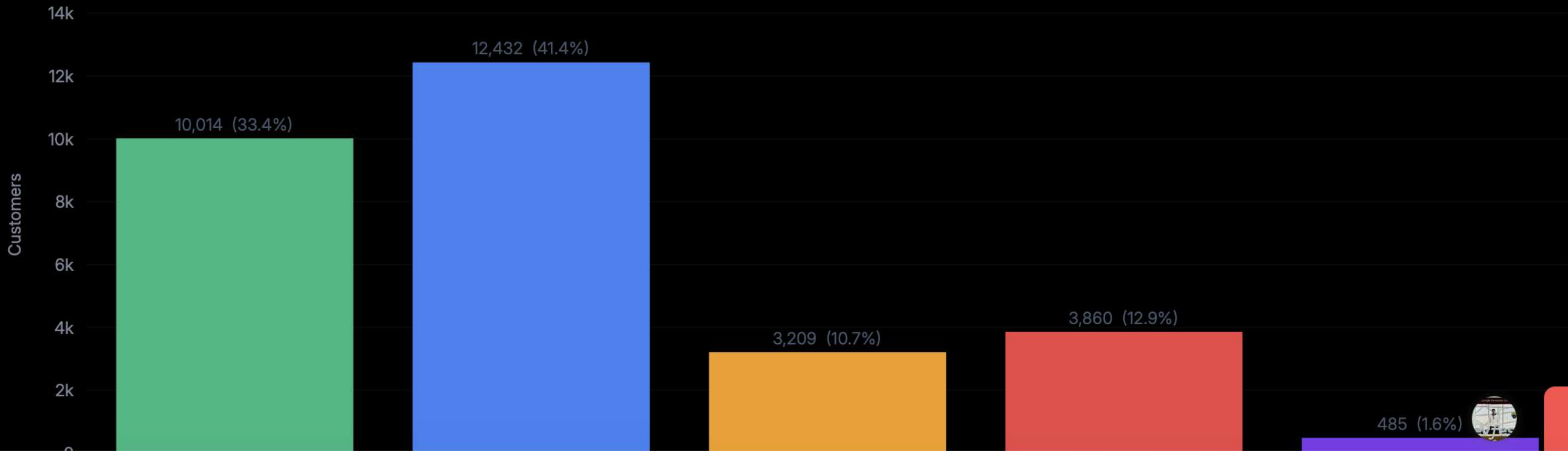
Fairness

What-If

Model

Recovery Tier Distribution

Final classification after B3.5 strategic default audit layer · 30,000 customers



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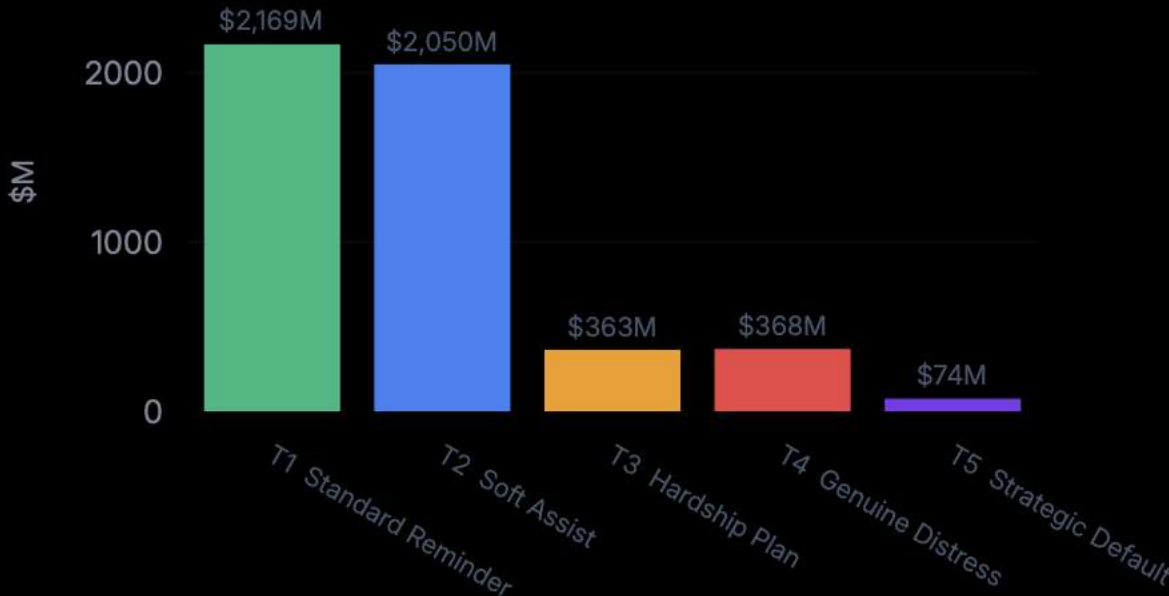
What-If

Model

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Credit Exposure by Tier

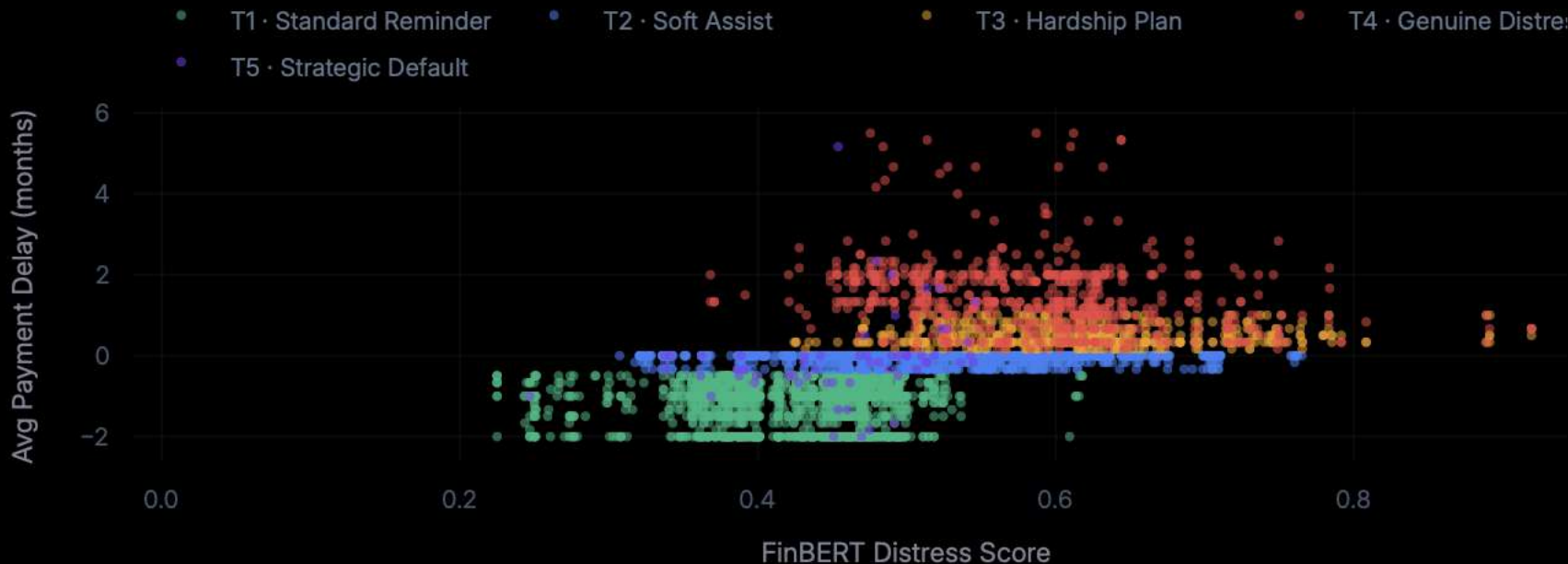
Total credit limit at risk · \$M



T4 + T5 exposure: \$442M of \$5.025M total

Risk Map — Distress vs Payment Delay

Each point = 1 customer · coloured by recovery tier · 5K sample



Select Customer

All 

SBDR_00000 

CUSTOMER

SBDR_00000

RECOVERY TIER

T2 · Soft Assist

NEXT BEST ACTION

Soft Assist SMS

AVG DISTRESS

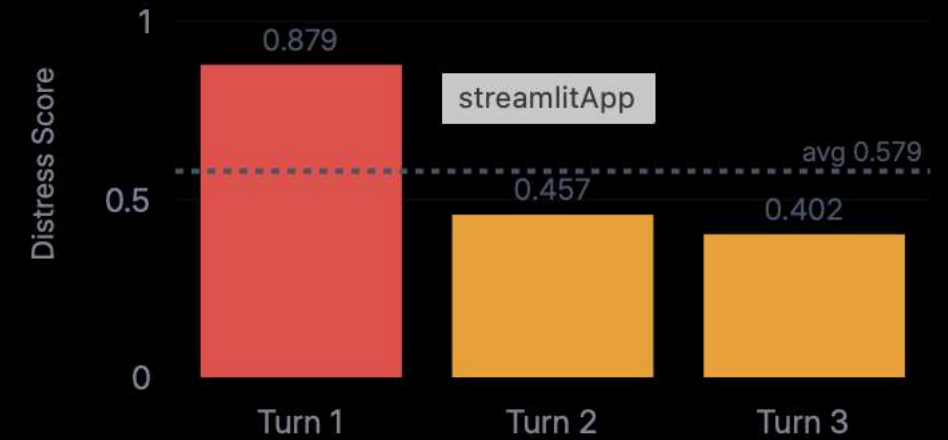
0.579

BILSTM ANOMALY

✓ Clear

Branch 1 · FinBERT

Sentiment distress score per chat turn



Sentiment shift T3→T1: ▼ 0.477 (de-escalating)

TURN 1 · 0.879

Branch 2 · BiLSTM

6-month payment sequence · anomaly detection



Reconstruction Error0.00221

Anomaly Flag✓ Clear

Branch 3 · XGBoost

Tier probability distribution

T1 · Standard Reminder	0.000
T2 · Soft Assist	0.999
T3 · Hardship Plan	0.000
T4 · Genuine Distress	0.000
T5 · Strategic Default	0.000

Sparkov Transaction Signals

Total Spend\$163,438.58

Monthly Avg74.53 



STRATEGIC DEFAULTS

485

Tier 5 confirmed

AUDIT ESCALATIONS

+266

Added by B3.5 layer

RULE A TRIGGERS

180

Label override

RULE C TRIGGERS

11

Fraud + anomaly + default

B3.5 Audit Spotlight

Case: SBDR_06150

Customer ID

SBDR_06150

Final Tier

T5 · Strategic Default

Distress Score

0.225

BiLSTM Anomaly

⚠ Detected

Avg Pay Delay

-0.83 mo

Fraud Rate

1.000

Default Label

Defaulted

Audit Rule

Rule A + C

Tier 5 — Strategic Default Confirmed

Legal referral required · All five audit conditions met

Rule A Engineered label override (default + anomaly + low distress)

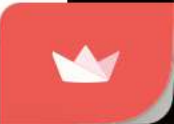
Strategic Default Roster

485 confirmed Tier 5 accounts

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Filter by customer ID

Customer	Distress	BiLSTM	Delay	Fraud	A
SBDR_00007	0.401⚠	-0.50	0.006	—	
SBDR_00065	0.400⚠	-0.50	0.012	✓	
SBDR_00108	0.432⚠	-0.50	0.015	—	
SBDR_00141	0.486⚠	-1.00	0.014	—	
SBDR_00232	0.474⚠	-0.67	0.005	—	
SBDR_00301	0.525⚠	-2.00	1.000	—	
SBDR_00389	0.511⚠	-0.67	0.007	✓	
SBDR_00522	0.484⚠	-0.67	0.007	✓	
SBDR_00529	0.472⚠	0.00	0.005	—	
SBDR_00645	0.484⚠	-1.17	0.005	—	
SBDR_00779	0.503⚠	-0.33	0.002	—	

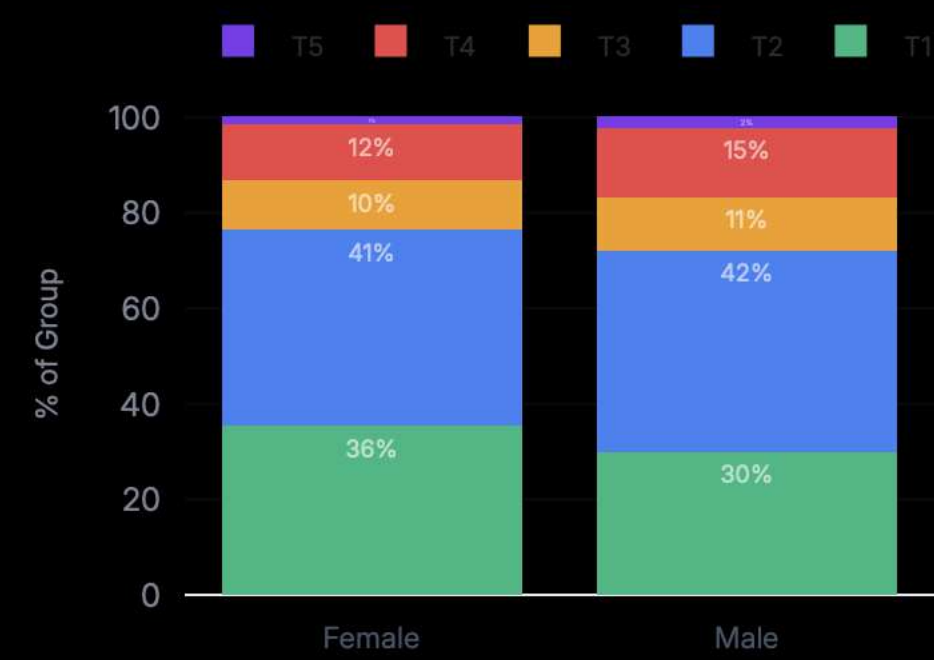


Demographic Fairness Audit (C2)

Tier 5 escalation rates are audited across sex, age group, and education level. SEX, AGE, and EDUCATION are not top SHAP drivers — escalation is driven by behavioural signals (avg_pay_delay, bilstm_anomaly_flag), not demographics.

By Gender

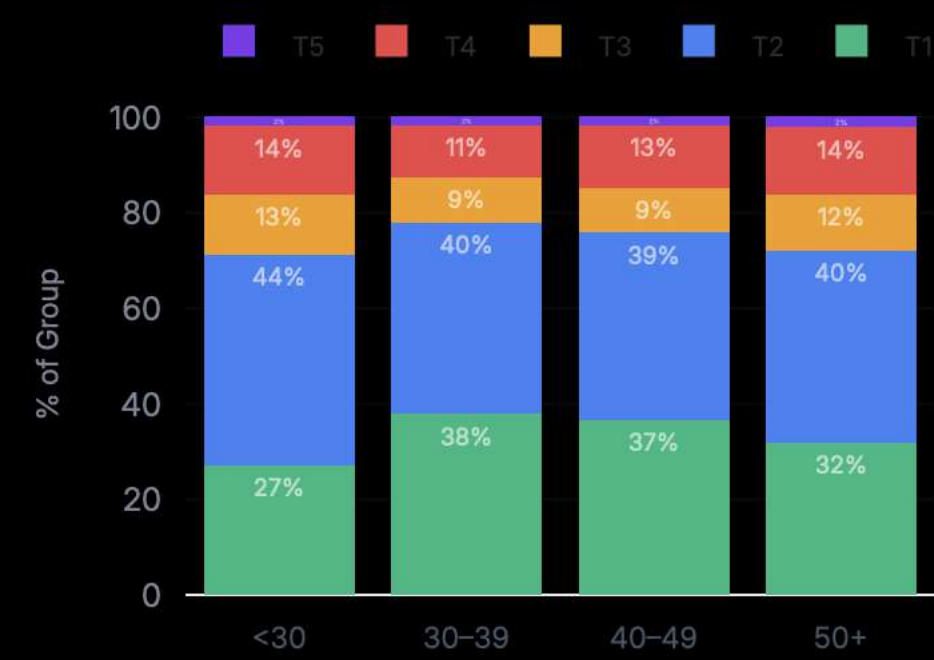
Stacked 100% bar — equal tier share = no gender bias



T5 gap — Male: 2.0% vs Female: 1.4% (0.7pp ✓ within tolerance)

By Age Group

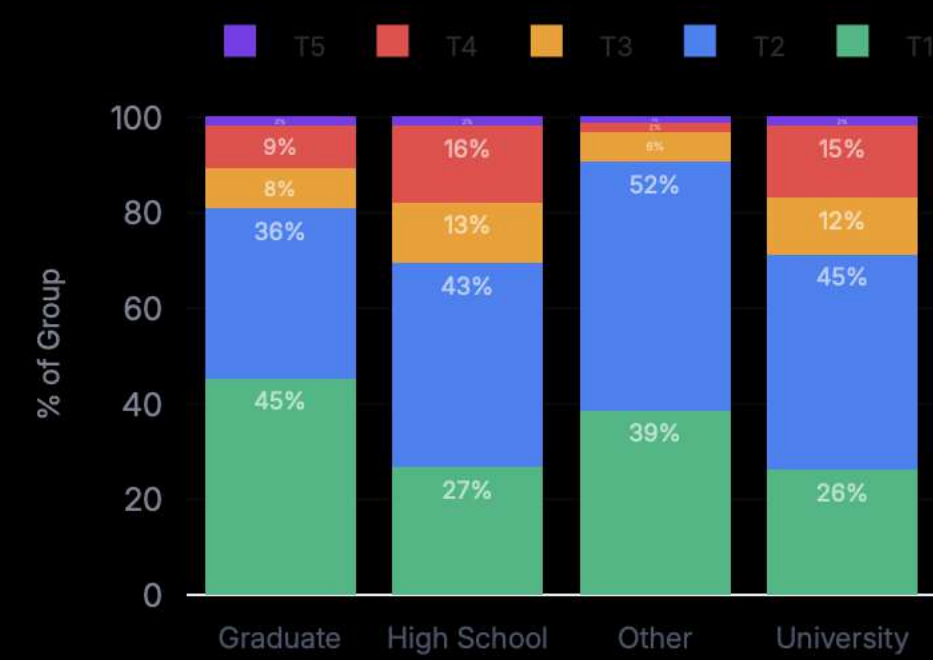
Binned into 4 cohorts — checks age-based escalation bias



Highest T5 rate: 50+ (2.0%)

By Education

1=Graduate · 2=University · 3=High School · 4=Other



Highest T5 rate: Graduate (1.7%)

TIER 5 ESCALATION RATE SUMMARY



Audit Verdict

No demographic group is disproportionately escalated to Tier 5



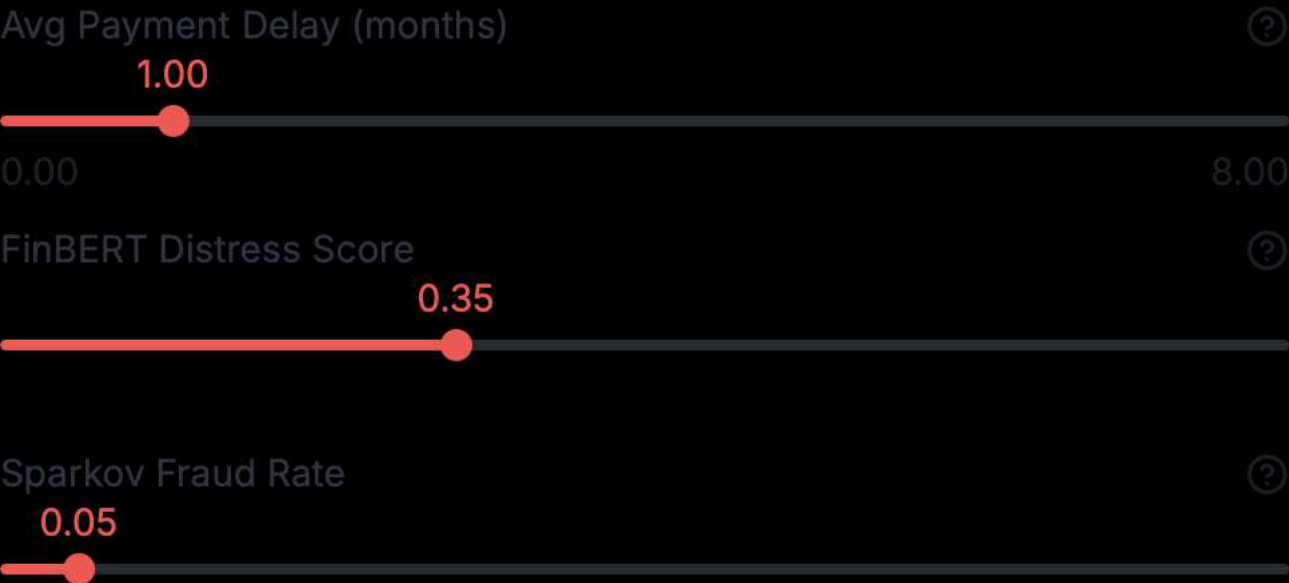
Recovery Tier Simulator

Adjust the customer signals below to see how the SBDR pipeline would classify them. The logic mirrors the B3.5 audit layer rules and XGBoost decision boundaries derived from SHAP analysis.
Top features: avg_pay_delay (SHAP 2.36) · bilstm_anomaly_flag (0.64) · distress_avg (0.14).

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Input Signals

Dial in the customer's key features



☐ BiLSTM Anomaly Detected

☐ Historical Default

Predicted Classification

Based on SBDR pipeline logic

RECOVERY TIER

T2

Soft Assist

NEXT BEST ACTION

Soft Assist SMS

Avg Payment Delay	1.0 mo
Distress Score	0.35
Fraud Rate	0.05
BiLSTM Anomaly	Clear
Default History	No

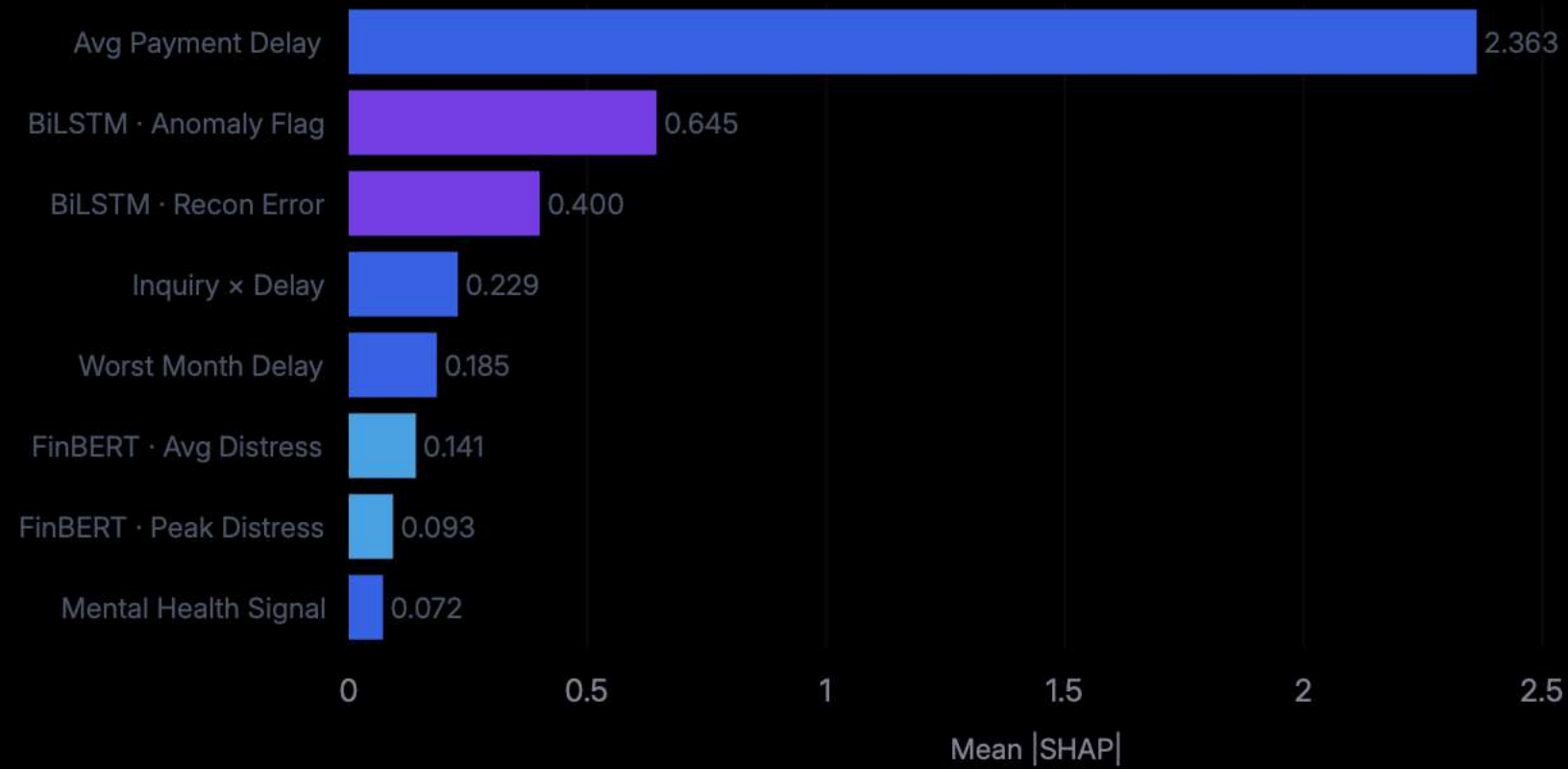




Top Predictive Features

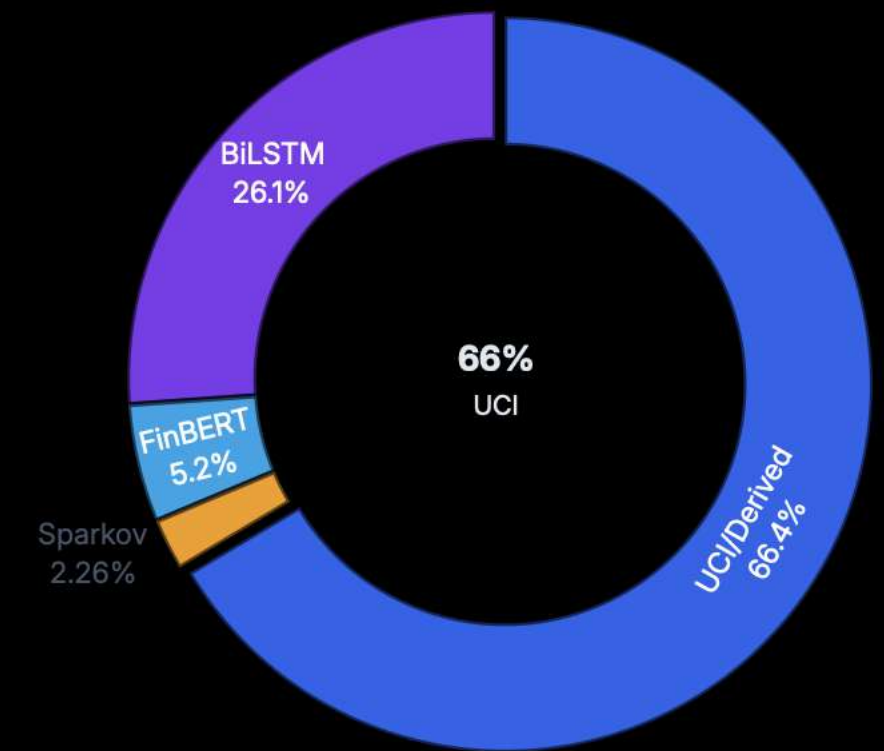
Mean |SHAP| value — coloured by AI branch

● UCI/Derived ● BiLSTM ● FinBERT ● Sparkov



AI Branch Contributions

Share of total predictive signal by model branch



Model Performance

XGBoost classifier · 116 features · 30K customers · 5 recovery tiers

ACCURACY

93.3%

Overall classification rate

AUC-ROC (MACRO)

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0.990

One-vs-rest discriminative power

F1 WEIGHTED

93.5%

Weighted by class frequency

F1 MACRO

76.1%

Unweighted across all 5 tiers

VAL LOG-LOSS

0.1301

Best early-stop validation loss

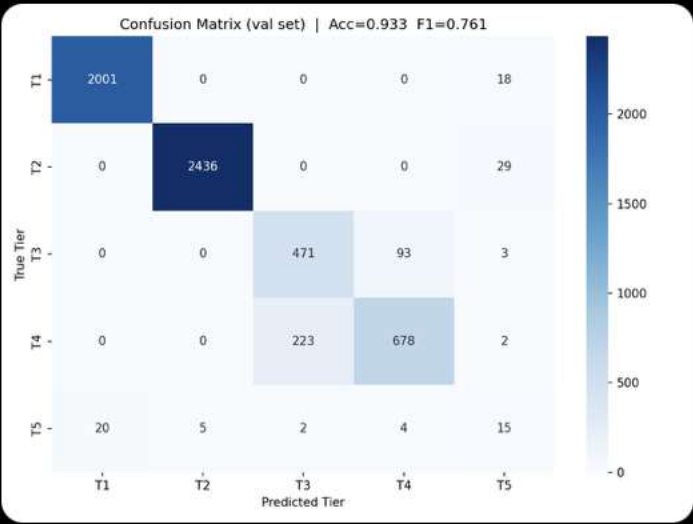
MODEL EVALUATION ARTIFACTS



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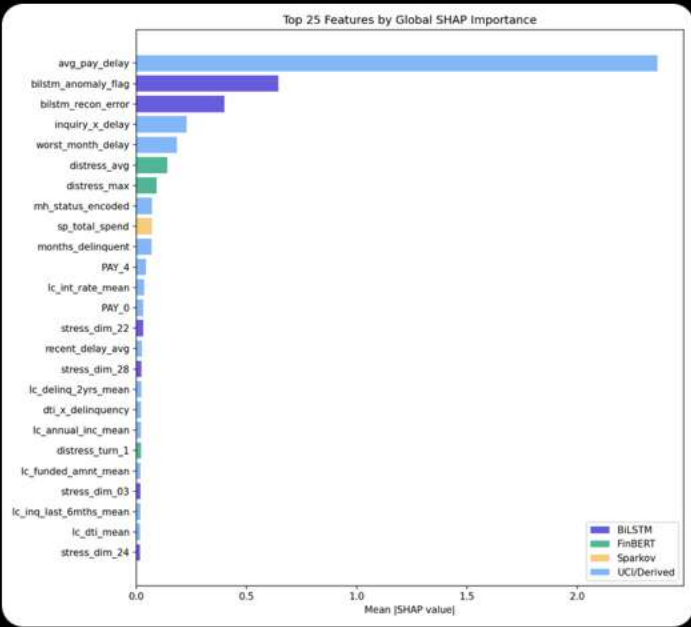
Confusion Matrix

Per-tier classification breakdown · XGBoost



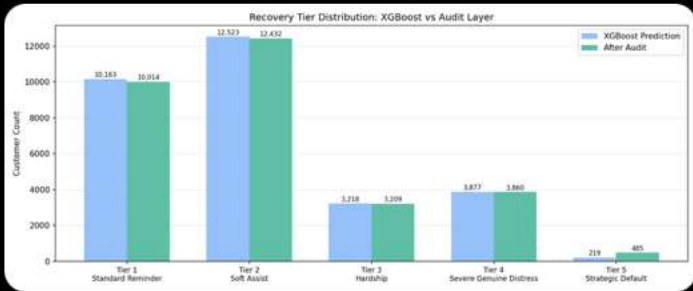
SHAP Feature Importance

Mean |SHAP| beeswarm from Notebook 08



B3.5 Audit Lift

XGBoost alone (219) vs after audit layer (485)



BiLSTM Training Loss

100-epoch convergence · best val loss 0.0015

